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 Analysis of Brain Tumor Using Machine Learning Algorithms

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Analysis of Brain Tumor Using Machine Learning Algorithms

Abhishek Kumar (22BEI0091)

School of Electrical Engineering

Vellore Institute of Technology

Vellore, Tamil Nadu, India

abhishek.kumar2022@vitstudent.ac.in

Snehil Raj (22BEE0079)

School of Electrical Engineering

Vellore Institute of Technology

Vellore, Tamil Nadu, India

snehil.raj2022@vitstudent.ac.in

Prof. Sankardoss V

Associate Professor Grade 1

School of Electrical Engineering

Vellore Institute of Technology

Vellore, Tamil Nadu, India

sankardoss.v@vit.ac.in

Abstract—Using magnetic resonance imaging (MRI) to diagnose brain tumours is a difficult and extremely sensitive clinical endeavour. The process of manually identifying tumours is laborious, subjective, and demands a high level of radiological skill. For automated multi-class tumour classification, this study suggests a hybrid approach that combines a Convolutional Neural Network (CNN) with conventional Machine Learning (ML) techniques like Support Vector Machine (SVM), Random Forest (RF), Logistic Regression, KNN, Naïve Bayes, Decision Tree, MLP, and XGBoost. 3788 MRI pictures classified as No Tumour, Glioma, Meningioma, and Pituitary Tumour made up the dataset. Comparative investigation reveals that ensemble machine learning models, such as Random Forest and XGBoost, outperform traditional classifiers in terms of accuracy and generalization. With an accuracy of 93%, the CNN model, however, performs noticeably better than all ML techniques, exhibiting a stronger capacity to extract textural and spatial characteristics. The findings show that deep learning is quite successful for classifying brain tumours using MRI, and it may be further enhanced with larger datasets and transfer learning.

Index Terms—Brain Tumor Detection, Machine Learning, Deep Learning, Convolutional Neural Networks, MRI Classification

I. INTRODUCTION

One of the most serious and potentially fatal neurological conditions, brain tumours affect millions of people annually all over the world. These cancers can spread from other regions of the body (metastatic tumours) or start in the brain (primary tumours). Gliomas, meningiomas, and pituitary tumours are the most prevalent kinds of primary brain tumours. Improving treatment success rates and patient survival requires early identification and precise categorization of these cancers. However, the size, location, morphology, and texture of brain tumours can vary greatly, making medical identification a very difficult and specialized procedure. Because of its excellent soft-tissue contrast, non-invasiveness, and multi-planar imaging capacity, magnetic resonance imaging (MRI) is commonly regarded as the gold standard for brain tumour screening. Even with these benefits, manually analyzing MRI images is still difficult. Radiologists have to pay close attention to minute changes in tissue density, shape, and intensity. In addition to being time-consuming, this kind of manual interpretation is vulnerable to subjectivity and diagnostic variability, particularly when

tumour borders are unclear or symptoms are vague. Traditional manual analysis techniques are no longer enough to guarantee quick and accurate clinical decision-making due to the exponential expansion of medical imaging data. Artificial Intelligence (AI) has been a game-changer in the healthcare sector in recent years, especially Machine Learning (ML) and Deep Learning (DL). Medical pictures have been subjected to machine learning algorithms to help in lesion identification, segmentation, and classification. However, the intricate spatial correlations found in MRI scans may not be fully captured by traditional machine learning algorithms, which rely largely on domain expertise and need handmade features. Convolutional Neural Networks (CNNs), in particular, have demonstrated impressive ability in automatically learning abstract and discriminative characteristics from pictures. Because CNNs can handle non-linear patterns, extract hierarchical spatial information, and generalize across many imaging variances, they are naturally well-suited for medical imaging tasks. CNNs perform better than conventional ML algorithms in terms of classification accuracy, robustness, and generalization, according to several studies. Comparing the effectiveness of machine learning models versus deep learning frameworks for multi-class brain tumour classification is still a major problem despite recent developments. There is a lack of thorough comparison evaluations because the majority of current research either just concentrate on ML models or DL architectures. In real-world medical settings, where computational resources, dataset size, and interpretability vary, choosing the best diagnostic tool requires an understanding of each approach's advantages and disadvantages. This study compares eight machine learning algorithms—Naïve Bayes, K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Support Vector Machine (SVM), Multilayer Perceptron (MLP), Random Forest, and XGBoost—with a specially created Convolutional Neural Network (CNN) in order to close these gaps. A dataset of 3788 MRI images from axial, coronal, and sagittal views, classified as No Tumour, Glioma, Meningioma, and Pituitary Tumour, was used to train and assess each model.

According to the results, the CNN model performs noticeably better because of its deep feature-learning capacity, even while conventional ML techniques attain respectable

accuracy. This demonstrates how deep learning has great potential to enhance automated brain tumour diagnosis in clinical procedures.

II. BACKGROUND AND RELATED WORK

Due to the intricacy of tumour heterogeneity and the therapeutic significance of early identification, brain tumour diagnosis and classification have long been major subjects in medical image analysis. Because of its improved soft-tissue contrast and capacity to detect anatomical and pathological alterations across many orientations, magnetic resonance imaging (MRI) has become the most dependable imaging technique for evaluating brain tumours. However, the use of automated, computer-assisted diagnostic methods is driven by the fact that manual evaluation of MRI images is still laborious and subject to inter-observer variability. Traditional Machine Learning (ML) methods were the main emphasis of early efforts, which mostly depended on manually created features including texture descriptors, intensity histograms, and morphological traits. Islam et al. (2013), for instance, used multifractal texture estimation for tumour diagnosis, showing the promise of texture-based analysis while pointing out feature quality sensitivity. El-Dahshan et al. (2014) demonstrated improvement over simple machine learning classifiers by combining Discrete Wavelet Transform (DWT) with Artificial Neural Networks (ANN), however they were still limited by manual feature engineering. In a similar vein, Huda et al. (2016) investigated hybrid feature selection using ensemble techniques to enhance classification on unbalanced medical data. The intricacy of MRI patterns and the requirement for expertly created features constrained the performance of these traditional machine learning models, notwithstanding their reasonable success. Researchers started using ensemble models, such Random Forest and XGBoost, which showed better resilience and generalization, to get over these restrictions. According to studies by Malarvizhi et al. (2022) and Vankdothu Hameed (2022), ensemble approaches might handle higher-dimensional features more successfully than solo classifiers. However, their capacity to discover deeper spatial structures present in MRI images was still limited by their reliance on pre-extracted or manually created features. An important turning point was the advent of Deep Learning (DL), especially Convolutional Neural Networks (CNNs). CNNs can capture complicated spatial and textural information without human involvement since they automatically develop hierarchical feature representations. CNN-based models considerably outperform traditional ML methods in tumour classification tasks, as shown by Seetha Raja (2018) and Abiwinanda et al. (2019). This was further supported by Badža Barjaktarović (2020), who used a rather shallow CNN architecture on MRI data to achieve good accuracy. Deep CNNs are superior for multi-class tumour classification, frequently exceeding 90% accuracy, according to more recent research (Khan et al., 2022; Saeedi et al., 2023). Despite these advancements, most existing studies evaluate either ML methods or CNNs separately. Few works provide a direct, comparative analysis of both approaches using the same dataset and evaluation

criteria. To address this gap, the present study compares eight widely used ML algorithms with a custom CNN model on a dataset of 3,788 MRI images, providing a unified analysis of their performance and suitability for multi-class brain tumour classification.

III. OBJECTIVES

Developing an effective and precise method for the automatic categorization of brain cancers using MRI images is the main goal of this project. The following particular goals were established in order to accomplish this:

- To assess how well various Machine Learning (ML) methods perform
For multi-class brain tumour classification, these methods include Naïve Bayes, K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Support Vector Machine (SVM), Multilayer Perceptron (MLP), Random Forest, and XGBoost.
- Convolutional Neural Network (CNN) design and implementation
ability to automatically extract textural and spatial characteristics from MRI pictures without requiring human feature engineering.
- To evaluate the performance of categorization between the CNN architecture and conventional ML models utilizing measures like accuracy, precision, recall, F1-score, and confusion matrices.
- To evaluate the advantages, disadvantages, and appropriateness of ML and DL methods for practical diagnostic applications, taking into account elements like generalization capacity, computational cost, and dataset complexity.
- To determine which model works best for dependable, automated multi-class brain tumour categorization to assist medical professionals in making precise and timely diagnoses.

IV. SYSTEM ARCHITECTURE

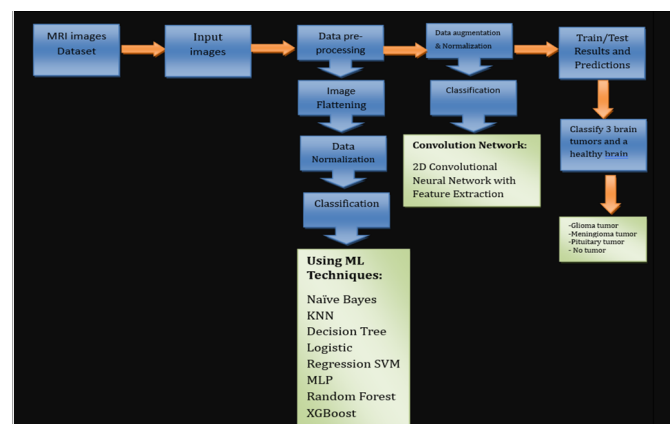


Fig. 1. Overall architecture for ML and CNN-based brain tumor classification.

Data collection, preprocessing, feature extraction, model training, and classification are the five main phases of the

system architecture for the Analysis of Brain Tumour Using Machine Learning Algorithms. The design combines a CNN-based deep learning pipeline with conventional machine learning models.

1) MRI Image Dataset

The system begins with a collection of MRI brain images belonging to four categories:

- Glioma
- Meningioma
- Pituitary Tumour
- No Tumour

These images include axial, coronal, and sagittal views.

2) Input Images

The raw MRI images are fed into the system for pre-processing and analysis.

3) Data Pre-processing

This stage prepares the images for both ML and CNN models through the following steps:

- **Image Flattening:** Converts 2D MRI images into 1D feature vectors for traditional ML models.
- **Data Normalization:** Scales pixel intensities for consistent model performance.
- **Data Augmentation (CNN only):** Includes rotation, flipping, zooming, and shifting to increase dataset diversity.

This creates two parallel streams: one for ML and one for CNN.

4) ML Pipeline

After flattening and normalization, the extracted feature vectors are passed to several machine learning classifiers:

- Naïve Bayes
- K-Nearest Neighbors (KNN)
- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)
- Multilayer Perceptron (MLP)
- Random Forest
- XGBoost

The ML models classify images based on predefined feature sets.

5) CNN Pipeline

The preprocessed images are also fed into the Convolutional Neural Network, which automatically extracts deep spatial and textural features:

- 2D Convolutional Layers
- Max-Pooling Layers
- Flattening Layer
- Dense Layers
- Softmax Output Layer

This pipeline eliminates the need for handcrafted features and provides more robust classification.

6) Train/Test Results and Predictions

Both pipelines (ML and CNN) generate predictions for each input MRI image. The evaluation metrics include:

- Accuracy
- Loss
- Precision
- Recall
- F1-score
- Confusion Matrices

7) Final Classification

The system classifies MRI images into one of the following four categories:

- Glioma Tumour
- Meningioma Tumour
- Pituitary Tumour
- No Tumour

The results demonstrate the superiority of the CNN model in multi-class brain tumour classification.

V. EXPERIMENTAL METHODOLOGY

A. Dataset

An organized pipeline comprising dataset preparation, pre-processing, feature extraction, model training, and performance evaluation makes up the experimental technique for this investigation. For multi-class brain tumour classification, both conventional Machine Learning (ML) techniques and a Convolutional Neural Network (CNN) were used. The actions taken are detailed below.

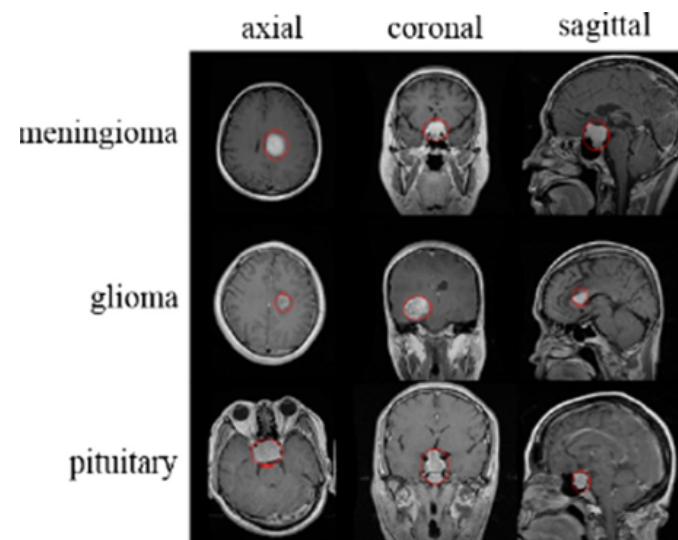


Fig. 2. Sample MRI dataset across four tumor categories.

1) Dataset Preparation

A dataset of 3788 MRI brain images was used, consisting of four classes:

- No Tumour
- Glioma Tumour
- Meningioma Tumour
- Pituitary Tumour

The dataset includes axial, coronal, and sagittal MRI views. An 80–20 train–test split ensured balanced representation across all classes.

2) Preprocessing

Preprocessing was performed to standardize the images for both ML and CNN pipelines:

- **Image Resizing:** All images resized to 224×224 pixels.
- **Normalization:** Pixel intensities scaled to the [0, 1] range.
- **Grayscale Conversion:** Reduces complexity and removes redundancy.
- **Data Augmentation (CNN only):** Applied to reduce overfitting:
 - Rotation
 - Horizontal/Vertical Flip
 - Zoom
 - Width/Height Shift
- **Flattening (ML only):** 2D MRI images flattened into 1D feature vectors.

3) Feature Extraction

Two feature extraction strategies were used:

- **ML Feature Extraction:** Handcrafted features such as:
 - Pixel intensity distributions
 - Texture features
 - Histogram features
 - Flattened pixel arrays
- **CNN Feature Learning:** Automatically extracted:
 - Low-level features (edges, textures)
 - Mid-level patterns (shapes, contours)
 - High-level spatial representations
 through stacked convolution and pooling layers.

4) Model Training

a) Machine Learning Models

The following ML models were trained using the extracted features:

- Naïve Bayes
- K-Nearest Neighbors (KNN)
- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)
- Multilayer Perceptron (MLP)
- Random Forest
- XGBoost

Default parameters were optimized using grid search where applicable.

b) CNN Model

A custom 2D CNN architecture was trained consisting of:

- Two Conv2D layers (32 and 64 filters)
- MaxPooling layers
- Flatten layer
- Dense layer (128 neurons)
- Softmax output (4 neurons)

Hyperparameters:

- **Optimizer:** Adam
- **Loss Function:** Sparse categorical crossentropy
- **Epochs:** 100
- **Batch Size:** 32

Training and validation accuracy/loss were monitored to track convergence and detect overfitting.

5) Model Evaluation

All models were evaluated using:

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix
- Training vs. Validation Curves (CNN only)

This enabled assessment of each model's generalization capability and diagnostic suitability.

6) Experimental Setup

All experiments were executed in Python using:

- TensorFlow/Keras for CNN
- Scikit-learn for ML models
- NumPy and Pandas for data handling
- Matplotlib and Seaborn for visualization

Execution Environment: Jupyter Notebook runtime

VI. RESULTS

Using a dataset of 3,788 MRI brain pictures, the effectiveness of eight machine learning (ML) methods and a proprietary convolutional neural network (CNN) was assessed. The algorithms were trained to categorize MRI data into four groups: pituitary tumour, meningioma, glioma, and no tumour. The outcomes show each approach's advantages, disadvantages, and relative efficacy.

A. Machine Learning Model Performance

1) *Naïve Bayes:* Out of all the models, the Naïve Bayes classifier performed the worst. There were high rates of misdiagnosis, particularly between meningioma and glioma tumours. Naïve Bayes is not appropriate for complicated MRI image classification because of its high independence requirements and limited capacity to collect spatial data. In summary, the baseline model for medical imaging tasks is subpar.

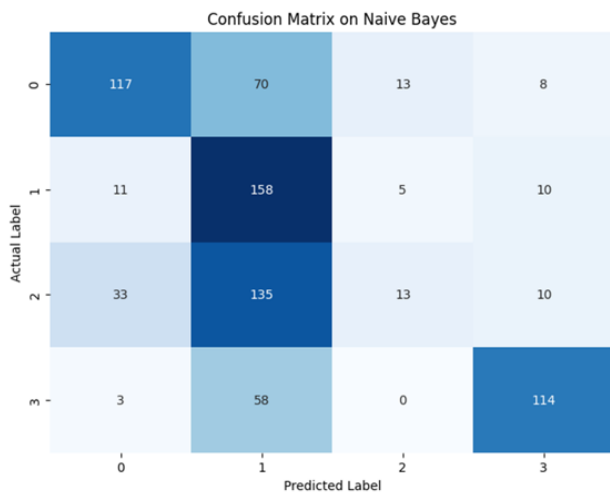


Fig. 3. Confusion Matrix – Naïve Bayes

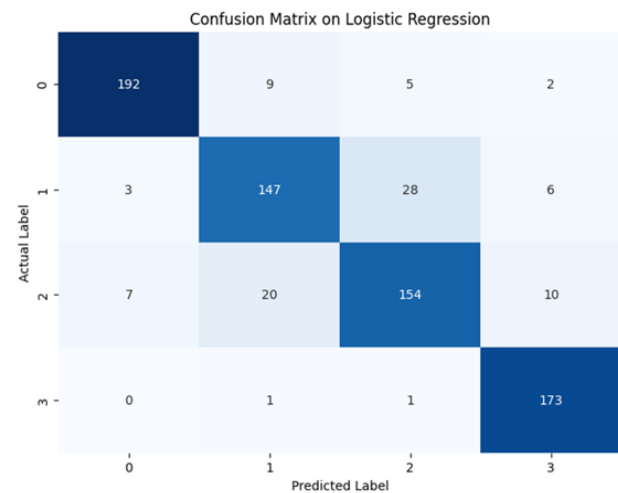


Fig. 5. Confusion Matrix – Logistic Regression

2) *KNN, or K-Nearest Neighbors*: KNN's performance was mediocre; it excelled in the Pituitary class but had trouble with MRI images with modest boundary changes. High-dimensional pixel data and noise sensitivity. In closely similar tumour types, misclassification increased. In conclusion, it functions satisfactorily but is unreliable for medical-grade precision.

4) *Decision Tree*: Although the Decision Tree classifier overfitted the training data, it did well on simpler classifications. Excellent precision for the No Tumour class. Due to overlapping pixel intensities, glioma and meningioma were significantly misclassified. Conclusion: Not the greatest option for MRI classification; interpretable but erratic.

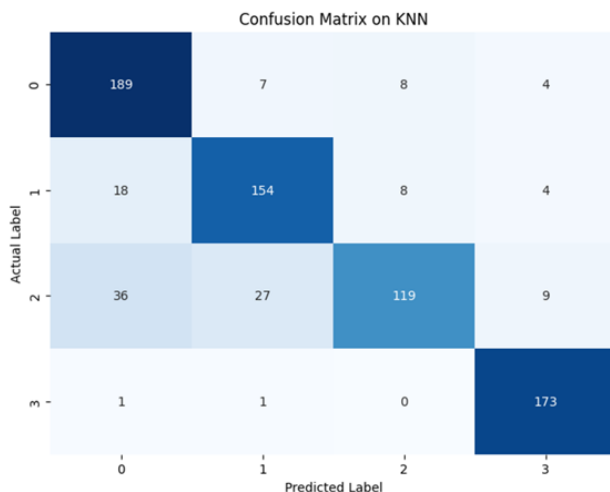


Fig. 4. Confusion Matrix – KNN

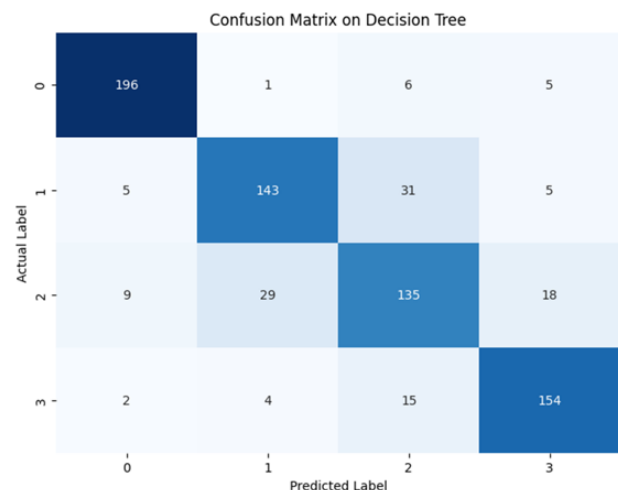


Fig. 6. Confusion Matrix – Decision Tree

3) *Logistic Regression*: On the No Tumour and Pituitary classes, Logistic Regression yielded respectable accuracy; however, it had trouble with Glioma and Meningioma. Its capacity to distinguish between intricate tumour textures is constrained by its linear decision boundary. In summary, the baseline model is good, but it is not adequate for multi-class tumour prediction.

5) *Support Vector Machine (SVM)*: SVM performed well and consistently in every category. Able to manage non-linear boundaries. Due to their similar physical appearance, misclassifications between gliomas and meningiomas were most common. Conclusion: Among the top-performing models for classical machine learning.

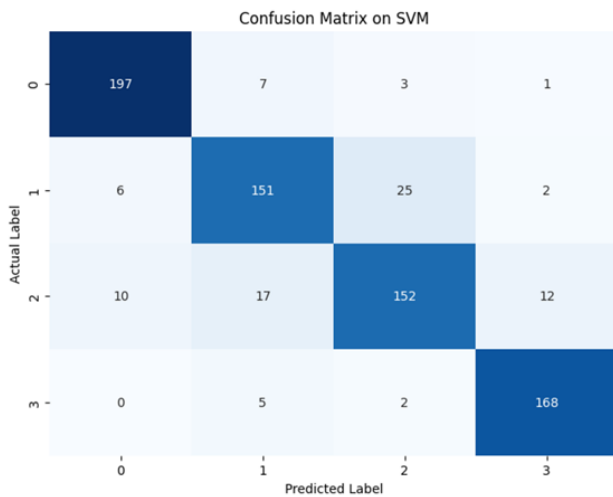


Fig. 7. Confusion Matrix – SVM

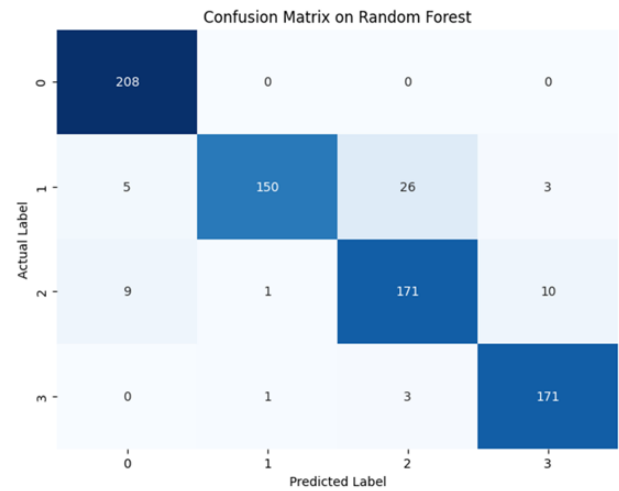


Fig. 9. Confusion Matrix – Random Forest

6) *Multilayer Perceptron (MLP)*: The MLP's performance was competitive, getting close to SVM accuracy. captured non-linear patterns with success. Better results compared to Decision Tree, KNN, and Naïve Bayes. Still, it's not as good as deep CNN feature extraction. In conclusion, the ML model is strong, although it is still not as good as CNN.

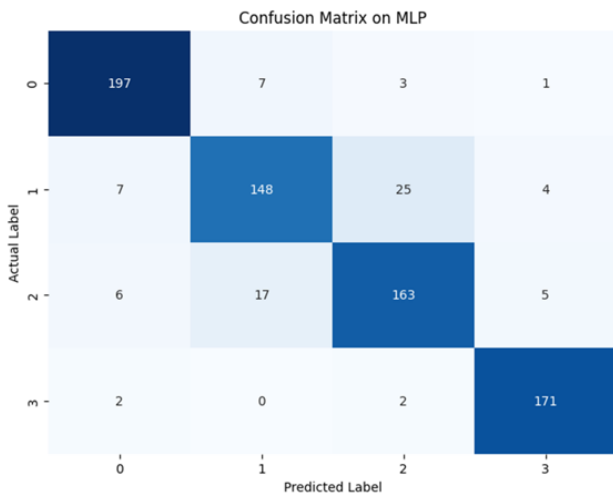


Fig. 8. Confusion Matrix – MLP

7) *Random Forest*: Random Forest outperformed the majority of traditional machine learning models. Stable generalization and high accuracy. Very little misclassification. Excellent results because of the combination of ensemble features. In conclusion, this is one of the most effective ML methods.

8) *XGBoost*: Among ML models, XGBoost demonstrated strong categorization across all tumour types, achieving the greatest accuracy. successfully manages non-linearities. Able to withstand overfitting. Excellent performance on high-dimensional feature sets. In conclusion, this machine learning model is the best overall and is very dependable for classifying tumours.

B. Convolutional Neural Network (CNN) Efficiency

When compared to all ML models, the CNN performed better, accomplishing: Training Precision: around 99%. Accuracy of Validation: 93%. Minimal overfitting and stable loss convergence.

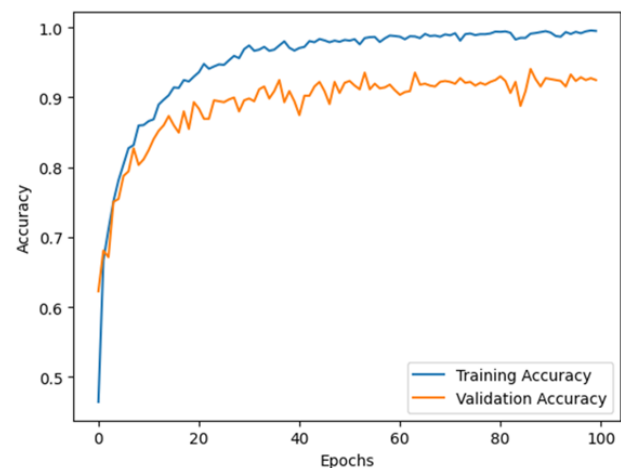


Fig. 10. CNN Training and Validation Accuracy

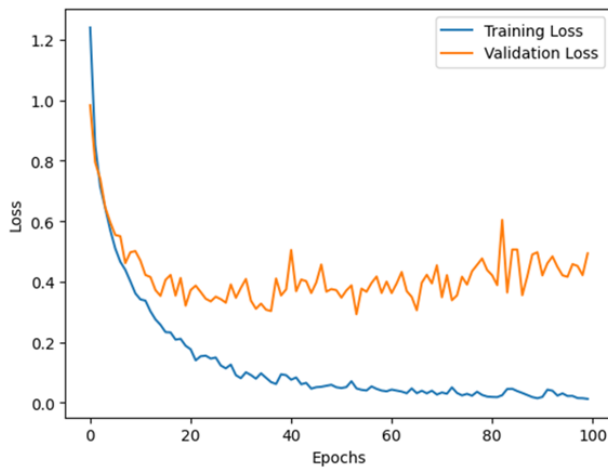


Fig. 11. CNN Training and Validation Loss

1) *Why CNN Did Better Than ML:* Multi-level characteristics (edges → textures → forms → tumour areas) are automatically learned. more resilient to changes in tumour form and MRI orientation. removes the need for hand-crafted, manual feature engineering.

2) *CNN Loss and Accuracy Trends:* For validation, accuracy grew gradually and reached a plateau at about 93%. Validation loss stabilized without divergence, whereas training loss became closer to zero. shows a highly predictive model that is well-generalized. Conclusion: Because CNN can directly learn hierarchical spatial patterns from MRI pictures, it obviously outperforms all ML techniques.

C. Model Performance Summary

TABLE I
MODEL PERFORMANCE SUMMARY

Naïve Bayes	Poor; cannot model image complexity
KNN	Moderate; sensitive to noise
Logistic Regression	Moderate; limited by linearity
Decision Tree	Good but overfitted
SVM	Very good; strong generalization
MLP	Very good; learns non-linear patterns
Random Forest	Excellent; robust ensemble
XGBoost	Outstanding; best ML performer
CNN	93% accuracy, best overall

The findings unequivocally show that deep learning outperforms conventional machine learning for MRI-based tumour categorisation. CNN is better at extracting and interpreting deep spatial patterns than Random Forest and XGBoost, despite their superior performance with designed features.

CNN's potential for incorporation into clinical decision-support systems is highlighted by its high accuracy (93%). CNN performance can reach radiologist-level accuracy with enough data and transfer learning.

All things considered, this comparison analysis confirms that CNNs are the best option for automated, multi-class brain tumour classification, but ML models can be quick, less complicated substitutes.

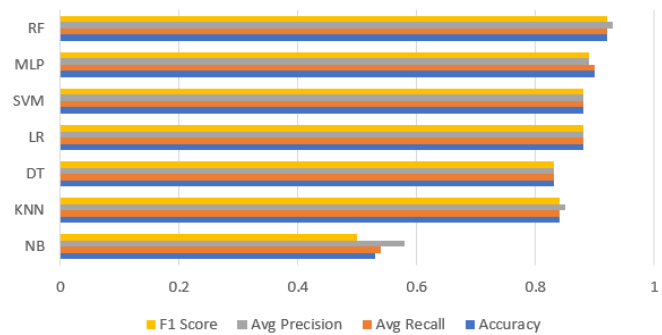


Fig. 12. Performance comparison of ML models showing Accuracy, Precision, Recall, and F1 Score across NB, KNN, DT, LR, SVM, MLP, and RF.

TABLE II
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	Avg Recall	Avg Precision	F1 Score
Naïve Bayes	0.53	0.54	0.58	0.50
K-Nearest Neighbors	0.84	0.84	0.85	0.84
Decision Tree	0.83	0.83	0.83	0.83
Logistic Regression	0.88	0.88	0.88	0.88
Support Vector Machine	0.88	0.88	0.88	0.88
Multilayer Perceptron	0.90	0.90	0.89	0.89
Random Forest	0.92	0.92	0.93	0.92

VII. DISCUSSION

Random Forest and XGBoost achieved the highest performance among ML models, due to their ability to combine multiple weak learners. Classical classifiers like Naïve Bayes and Logistic Regression performed poorly due to MRI complexity.

The CNN model outperformed all ML models with a validation accuracy of 93%, thanks to its ability to extract deep spatial features without manual engineering. CNN loss curves showed stable convergence, indicating effective generalization.

Overall, CNN is the most robust approach, while XGBoost provides a fast and lightweight alternative.

VIII. CONCLUSION

Eight machine learning techniques and a convolutional neural network (CNN) for multi-class brain tumour classification using MRI data were thoroughly compared in this study. On a dataset of 3,788 MRI scans, a bespoke CNN architecture was compared against traditional machine learning models such as Naïve Bayes, KNN, Logistic Regression, Decision Tree, SVM, MLP, Random Forest, and XGBoost. The findings showed that even though a number of machine learning models—Random Forest and XGBoost in particular—performed well, they were still constrained by their reliance on manually created features and their incapacity to completely capture the intricate spatial patterns found in MRI scans. On the other hand, because of its deep feature-learning capabilities, the CNN model demonstrated greater generalization and achieved a validation accuracy of 93%, outperforming all ML techniques. In order to distinguish between gliomas, meningiomas, pituitary tumours,

and normal brain scans more accurately, the CNN successfully learnt hierarchical spatial and textural properties straight from raw MRI images.

Overall, the results show that the most dependable and scalable method for automated brain tumour classification is deep learning, especially CNN-based architectures. CNN-based systems offer great potential to assist radiologists in early diagnosis and improve clinical decision-making in actual medical settings with additional advancements like bigger datasets, transfer learning, and integration of sophisticated architectures like ResNet or EfficientNet.

IX. ACKNOWLEDGMENT

We, Abhishek Kumar (22BEI0091) and Snehil Raj (22BEE0079), would want to sincerely thank everyone who helped us with our research, "Analysis of Brain Tumour Using Machine Learning Algorithms." We express our sincere gratitude to our mentor, Prof. Sankardoss V, Associate Professor Grade 1, School of Electrical Engineering, for his constant direction, perceptive criticism, and support during this project. His knowledge and guidance have been crucial in determining the course and calibre of our study. Additionally, we would like to express our gratitude to the School of Electrical Engineering at Vellore Institute of Technology for providing the tools and a supportive learning environment needed to complete this project. Lastly, we would like to thank everyone who helped us with this effort, whether directly or indirectly, and the publicly accessible MRI datasets. Their assistance has been crucial to this research's effective conclusion.

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Analysis of Brain Tumor Using Machine Learning Algorithms

Abhishek Kumar (22BEI0091)

School of Electrical Engineering

Vellore Institute of Technology

Vellore, Tamil Nadu, India

abhishek.kumar2022@vitstudent.ac.in

Snehil Raj (22BEE0079)

School of Electrical Engineering

Vellore Institute of Technology

Vellore, Tamil Nadu, India

snehil.raj2022@vitstudent.ac.in

Prof. Sankardoss V

Associate Professor Grade 1

School of Electrical Engineering

Vellore Institute of Technology

Vellore, Tamil Nadu, India

sankardoss.v@vit.ac.in

Abstract—Using magnetic resonance imaging (MRI) to diagnose brain tumours is a difficult and extremely sensitive clinical endeavour. The process of manually identifying tumours is laborious, subjective, and demands a high level of radiological skill. For automated multi-class tumour classification, this study suggests a hybrid approach that combines a Convolutional Neural Network (CNN) with conventional Machine Learning (ML) techniques like Support Vector Machine (SVM), Random Forest (RF), Logistic Regression, KNN, Naïve Bayes, Decision Tree, MLP, and XGBoost. 3788 MRI pictures classified as No Tumour, Glioma, Meningioma, and Pituitary Tumour made up the dataset. Comparative investigation reveals that ensemble machine learning models, such as Random Forest and XGBoost, outperform traditional classifiers in terms of accuracy and generalization. With an accuracy of 93%, the CNN model, however, performs noticeably better than all ML techniques, exhibiting a stronger capacity to extract textural and spatial characteristics. The findings show that deep learning is quite successful for classifying brain tumours using MRI, and it may be further enhanced with larger datasets and transfer learning.

Index Terms—Brain Tumor Detection, Machine Learning, Deep Learning, Convolutional Neural Networks, MRI Classification

I. INTRODUCTION

One of the most serious and potentially fatal neurological conditions, brain tumours affect millions of people annually all over the world. These cancers can spread from other regions of the body (metastatic tumours) or start in the brain (primary tumours). Gliomas, meningiomas, and pituitary tumours are the most prevalent kinds of primary brain tumours. Improving treatment success rates and patient survival requires early identification and precise categorization of these cancers. However, the size, location, morphology, and texture of brain tumours can vary greatly, making medical identification a very difficult and specialized procedure. Because of its excellent soft-tissue contrast, non-invasiveness, and multi-planar imaging capacity, magnetic resonance imaging (MRI) is commonly regarded as the gold standard for brain tumour screening. Even with these benefits, manually analyzing MRI images is still difficult. Radiologists have to pay close attention to minute changes in tissue density, shape, and intensity. In addition to being time-consuming, this kind of manual interpretation is vulnerable to subjectivity and diagnostic variability, particularly when

tumour borders are unclear or symptoms are vague. Traditional manual analysis techniques are no longer enough to guarantee quick and accurate clinical decision-making due to the exponential expansion of medical imaging data. Artificial Intelligence (AI) has been a game-changer in the healthcare sector in recent years, especially Machine Learning (ML) and Deep Learning (DL). Medical pictures have been subjected to machine learning algorithms to help in lesion identification, segmentation, and classification. However, the intricate spatial correlations found in MRI scans may not be fully captured by traditional machine learning algorithms, which rely largely on domain expertise and need handmade features. Convolutional Neural Networks (CNNs), in particular, have demonstrated impressive ability in automatically learning abstract and discriminative characteristics from pictures. Because CNNs can handle non-linear patterns, extract hierarchical spatial information, and generalize across many imaging variances, they are naturally well-suited for medical imaging tasks. CNNs perform better than conventional ML algorithms in terms of classification accuracy, robustness, and generalization, according to several studies. Comparing the effectiveness of machine learning models versus deep learning frameworks for multi-class brain tumour classification is still a major problem despite recent developments. There is a lack of thorough comparison evaluations because the majority of current research either just concentrate on ML models or DL architectures. In real-world medical settings, where computational resources, dataset size, and interpretability vary, choosing the best diagnostic tool requires an understanding of each approach's advantages and disadvantages. This study compares eight machine learning algorithms—Naïve Bayes, K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Support Vector Machine (SVM), Multilayer Perceptron (MLP), Random Forest, and XGBoost—with a specially created Convolutional Neural Network (CNN) in order to close these gaps. A dataset of 3788 MRI images from axial, coronal, and sagittal views, classified as No Tumour, Glioma, Meningioma, and Pituitary Tumour, was used to train and assess each model.

According to the results, the CNN model performs noticeably better because of its deep feature-learning capacity, even while conventional ML techniques attain respectable

accuracy. This demonstrates how deep learning has great potential to enhance automated brain tumour diagnosis in clinical procedures.

II. BACKGROUND AND RELATED WORK

Due to the intricacy of tumour heterogeneity and the therapeutic significance of early identification, brain tumour diagnosis and classification have long been major subjects in medical image analysis. Because of its improved soft-tissue contrast and capacity to detect anatomical and pathological alterations across many orientations, magnetic resonance imaging (MRI) has become the most dependable imaging technique for evaluating brain tumours. However, the use of automated, computer-assisted diagnostic methods is driven by the fact that manual evaluation of MRI images is still laborious and subject to inter-observer variability. Traditional Machine Learning (ML) methods were the main emphasis of early efforts, which mostly depended on manually created features including texture descriptors, intensity histograms, and morphological traits. Islam et al. (2013), for instance, used multifractal texture estimation for tumour diagnosis, showing the promise of texture-based analysis while pointing out feature quality sensitivity. El-Dahshan et al. (2014) demonstrated improvement over simple machine learning classifiers by combining Discrete Wavelet Transform (DWT) with Artificial Neural Networks (ANN), however they were still limited by manual feature engineering. In a similar vein, Huda et al. (2016) investigated hybrid feature selection using ensemble techniques to enhance classification on unbalanced medical data. The intricacy of MRI patterns and the requirement for expertly created features constrained the performance of these traditional machine learning models, notwithstanding their reasonable success. Researchers started using ensemble models, such Random Forest and XGBoost, which showed better resilience and generalization, to get over these restrictions. According to studies by Malarvizhi et al. (2022) and Vankdothu Hameed (2022), ensemble approaches might handle higher-dimensional features more successfully than solo classifiers. However, their capacity to discover deeper spatial structures present in MRI images was still limited by their reliance on pre-extracted or manually created features. An important turning point was the advent of Deep Learning (DL), especially Convolutional Neural Networks (CNNs). CNNs can capture complicated spatial and textural information without human involvement since they automatically develop hierarchical feature representations. CNN-based models considerably outperform traditional ML methods in tumour classification tasks, as shown by Seetha Raja (2018) and Abiwinanda et al. (2019). This was further supported by Badža Barjaktarović (2020), who used a rather shallow CNN architecture on MRI data to achieve good accuracy. Deep CNNs are superior for multi-class tumour classification, frequently exceeding 90% accuracy, according to more recent research (Khan et al., 2022; Saeedi et al., 2023). Despite these advancements, most existing studies evaluate either ML methods or CNNs separately. Few works provide a direct, comparative analysis of both approaches using the same dataset and evaluation

criteria. To address this gap, the present study compares eight widely used ML algorithms with a custom CNN model on a dataset of 3,788 MRI images, providing a unified analysis of their performance and suitability for multi-class brain tumour classification.

III. OBJECTIVES

Developing an effective and precise method for the automatic categorization of brain cancers using MRI images is the main goal of this project. The following particular goals were established in order to accomplish this:

- To assess how well various Machine Learning (ML) methods perform
For multi-class brain tumour classification, these methods include Naïve Bayes, K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Support Vector Machine (SVM), Multilayer Perceptron (MLP), Random Forest, and XGBoost.
- Convolutional Neural Network (CNN) design and implementation
ability to automatically extract textural and spatial characteristics from MRI pictures without requiring human feature engineering.
- To evaluate the performance of categorization between the CNN architecture and conventional ML models utilizing measures like accuracy, precision, recall, F1-score, and confusion matrices.
- To evaluate the advantages, disadvantages, and appropriateness of ML and DL methods for practical diagnostic applications, taking into account elements like generalization capacity, computational cost, and dataset complexity.
- To determine which model works best for dependable, automated multi-class brain tumour categorization to assist medical professionals in making precise and timely diagnoses.

IV. SYSTEM ARCHITECTURE

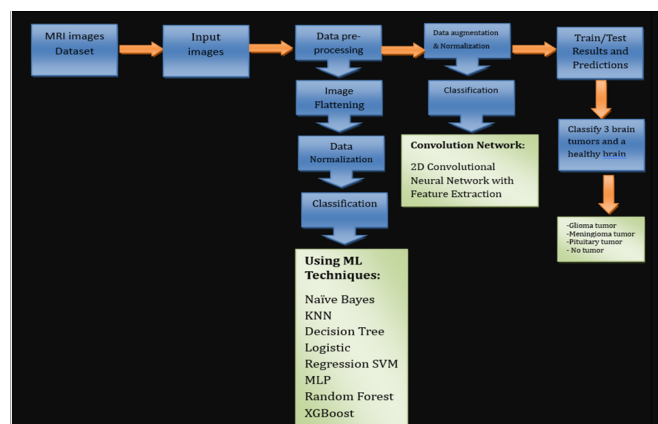


Fig. 1. Overall architecture for ML and CNN-based brain tumor classification.

Data collection, preprocessing, feature extraction, model training, and classification are the five main phases of the

system architecture for the Analysis of Brain Tumour Using Machine Learning Algorithms. The design combines a CNN-based deep learning pipeline with conventional machine learning models.

1) MRI Image Dataset

The system begins with a collection of MRI brain images belonging to four categories:

- Glioma
- Meningioma
- Pituitary Tumour
- No Tumour

These images include axial, coronal, and sagittal views.

2) Input Images

The raw MRI images are fed into the system for pre-processing and analysis.

3) Data Pre-processing

This stage prepares the images for both ML and CNN models through the following steps:

- **Image Flattening:** Converts 2D MRI images into 1D feature vectors for traditional ML models.
- **Data Normalization:** Scales pixel intensities for consistent model performance.
- **Data Augmentation (CNN only):** Includes rotation, flipping, zooming, and shifting to increase dataset diversity.

This creates two parallel streams: one for ML and one for CNN.

4) ML Pipeline

After flattening and normalization, the extracted feature vectors are passed to several machine learning classifiers:

- Naïve Bayes
- K-Nearest Neighbors (KNN)
- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)
- Multilayer Perceptron (MLP)
- Random Forest
- XGBoost

The ML models classify images based on predefined feature sets.

5) CNN Pipeline

The preprocessed images are also fed into the Convolutional Neural Network, which automatically extracts deep spatial and textural features:

- 2D Convolutional Layers
- Max-Pooling Layers
- Flattening Layer
- Dense Layers
- Softmax Output Layer

This pipeline eliminates the need for handcrafted features and provides more robust classification.

6) Train/Test Results and Predictions

Both pipelines (ML and CNN) generate predictions for each input MRI image. The evaluation metrics include:

- Accuracy
- Loss
- Precision
- Recall
- F1-score
- Confusion Matrices

7) Final Classification

The system classifies MRI images into one of the following four categories:

- Glioma Tumour
- Meningioma Tumour
- Pituitary Tumour
- No Tumour

The results demonstrate the superiority of the CNN model in multi-class brain tumour classification.

V. EXPERIMENTAL METHODOLOGY

A. Dataset

An organized pipeline comprising dataset preparation, pre-processing, feature extraction, model training, and performance evaluation makes up the experimental technique for this investigation. For multi-class brain tumour classification, both conventional Machine Learning (ML) techniques and a Convolutional Neural Network (CNN) were used. The actions taken are detailed below.

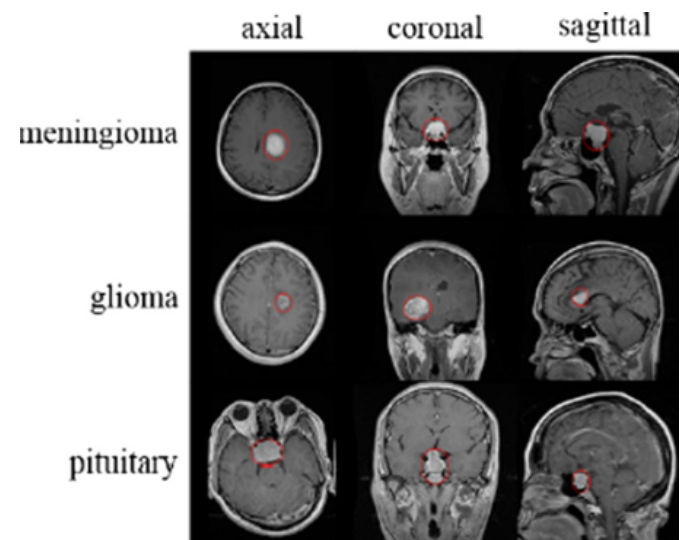


Fig. 2. Sample MRI dataset across four tumor categories.

1) Dataset Preparation

A dataset of 3788 MRI brain images was used, consisting of four classes:

- No Tumour
- Glioma Tumour
- Meningioma Tumour
- Pituitary Tumour

The dataset includes axial, coronal, and sagittal MRI views. An 80–20 train–test split ensured balanced representation across all classes.

2) Preprocessing

Preprocessing was performed to standardize the images for both ML and CNN pipelines:

- **Image Resizing:** All images resized to 224×224 pixels.
- **Normalization:** Pixel intensities scaled to the [0, 1] range.
- **Grayscale Conversion:** Reduces complexity and removes redundancy.
- **Data Augmentation (CNN only):** Applied to reduce overfitting:
 - Rotation
 - Horizontal/Vertical Flip
 - Zoom
 - Width/Height Shift
- **Flattening (ML only):** 2D MRI images flattened into 1D feature vectors.

3) Feature Extraction

Two feature extraction strategies were used:

- **ML Feature Extraction:** Handcrafted features such as:
 - Pixel intensity distributions
 - Texture features
 - Histogram features
 - Flattened pixel arrays
- **CNN Feature Learning:** Automatically extracted:
 - Low-level features (edges, textures)
 - Mid-level patterns (shapes, contours)
 - High-level spatial representations
 through stacked convolution and pooling layers.

4) Model Training

a) Machine Learning Models

The following ML models were trained using the extracted features:

- Naïve Bayes
- K-Nearest Neighbors (KNN)
- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)
- Multilayer Perceptron (MLP)
- Random Forest
- XGBoost

Default parameters were optimized using grid search where applicable.

b) CNN Model

A custom 2D CNN architecture was trained consisting of:

- Two Conv2D layers (32 and 64 filters)
- MaxPooling layers
- Flatten layer
- Dense layer (128 neurons)
- Softmax output (4 neurons)

Hyperparameters:

- **Optimizer:** Adam
- **Loss Function:** Sparse categorical crossentropy
- **Epochs:** 100
- **Batch Size:** 32

Training and validation accuracy/loss were monitored to track convergence and detect overfitting.

5) Model Evaluation

All models were evaluated using:

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix
- Training vs. Validation Curves (CNN only)

This enabled assessment of each model's generalization capability and diagnostic suitability.

6) Experimental Setup

All experiments were executed in Python using:

- TensorFlow/Keras for CNN
- Scikit-learn for ML models
- NumPy and Pandas for data handling
- Matplotlib and Seaborn for visualization

Execution Environment: Jupyter Notebook runtime

VI. RESULTS

Using a dataset of 3,788 MRI brain pictures, the effectiveness of eight machine learning (ML) methods and a proprietary convolutional neural network (CNN) was assessed. The algorithms were trained to categorize MRI data into four groups: pituitary tumour, meningioma, glioma, and no tumour. The outcomes show each approach's advantages, disadvantages, and relative efficacy.

A. Machine Learning Model Performance

1) *Naïve Bayes:* Out of all the models, the Naïve Bayes classifier performed the worst. There were high rates of misdiagnosis, particularly between meningioma and glioma tumours. Naïve Bayes is not appropriate for complicated MRI image classification because of its high independence requirements and limited capacity to collect spatial data. In summary, the baseline model for medical imaging tasks is subpar.

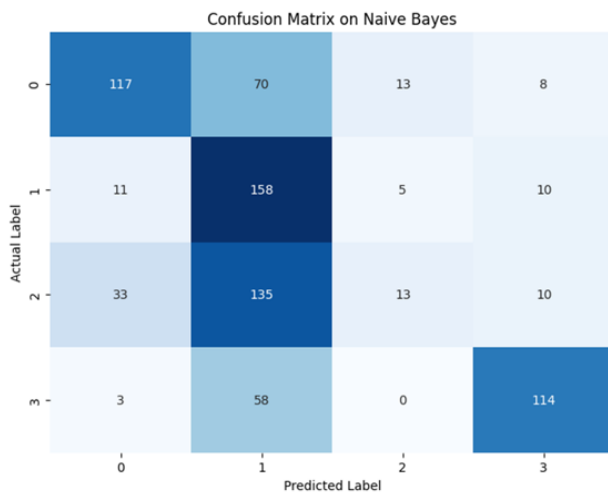


Fig. 3. Confusion Matrix – Naïve Bayes

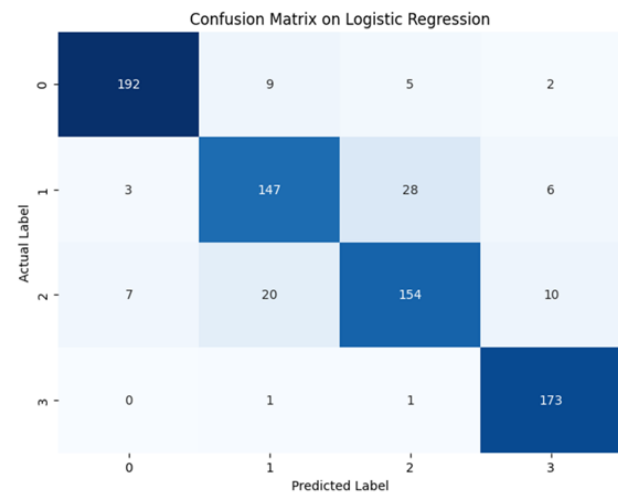


Fig. 5. Confusion Matrix – Logistic Regression

2) *KNN, or K-Nearest Neighbors*: KNN's performance was mediocre; it excelled in the Pituitary class but had trouble with MRI images with modest boundary changes. High-dimensional pixel data and noise sensitivity. In closely similar tumour types, misclassification increased. In conclusion, it functions satisfactorily but is unreliable for medical-grade precision.

4) *Decision Tree*: Although the Decision Tree classifier overfitted the training data, it did well on simpler classifications. Excellent precision for the No Tumour class. Due to overlapping pixel intensities, glioma and meningioma were significantly misclassified. Conclusion: Not the greatest option for MRI classification; interpretable but erratic.

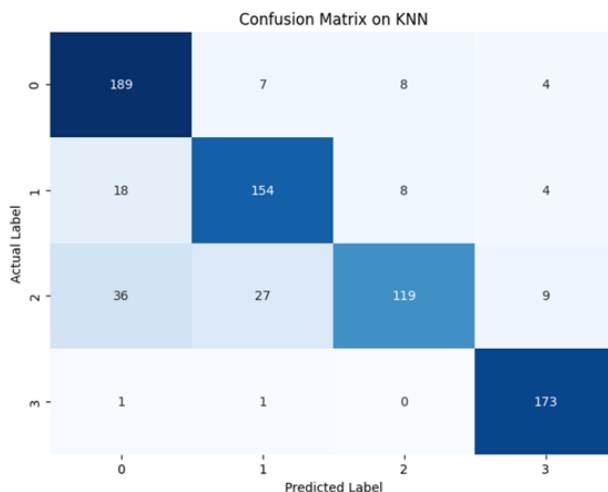


Fig. 4. Confusion Matrix – KNN

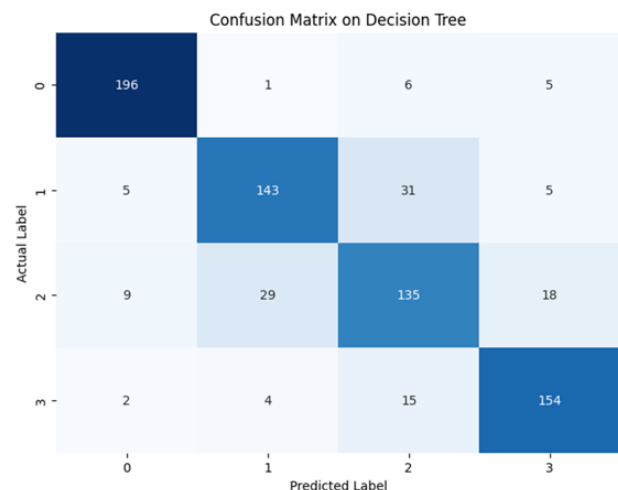


Fig. 6. Confusion Matrix – Decision Tree

3) *Logistic Regression*: On the No Tumour and Pituitary classes, Logistic Regression yielded respectable accuracy; however, it had trouble with Glioma and Meningioma. Its capacity to distinguish between intricate tumour textures is constrained by its linear decision boundary. In summary, the baseline model is good, but it is not adequate for multi-class tumour prediction.

5) *Support Vector Machine (SVM)*: SVM performed well and consistently in every category. Able to manage non-linear boundaries. Due to their similar physical appearance, misclassifications between gliomas and meningiomas were most common. Conclusion: Among the top-performing models for classical machine learning.

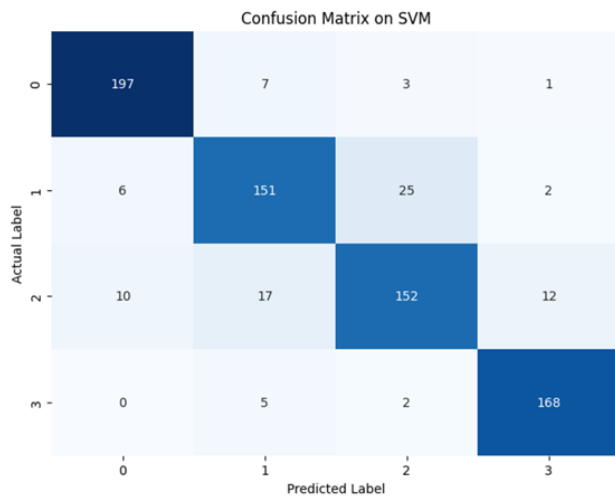


Fig. 7. Confusion Matrix – SVM

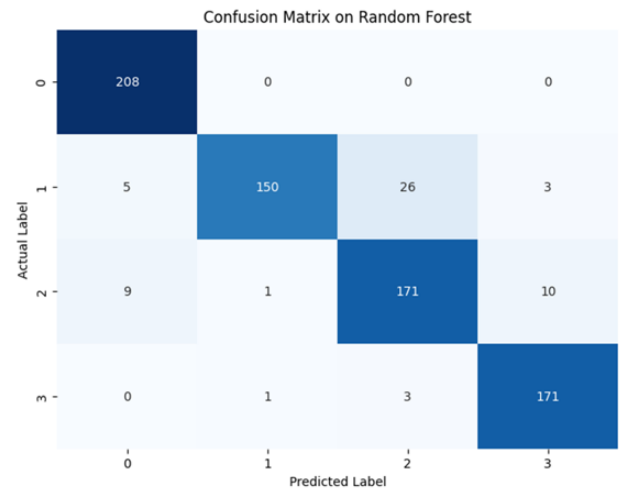


Fig. 9. Confusion Matrix – Random Forest

6) *Multilayer Perceptron (MLP)*: The MLP's performance was competitive, getting close to SVM accuracy. captured non-linear patterns with success. Better results compared to Decision Tree, KNN, and Naïve Bayes. Still, it's not as good as deep CNN feature extraction. In conclusion, the ML model is strong, although it is still not as good as CNN.

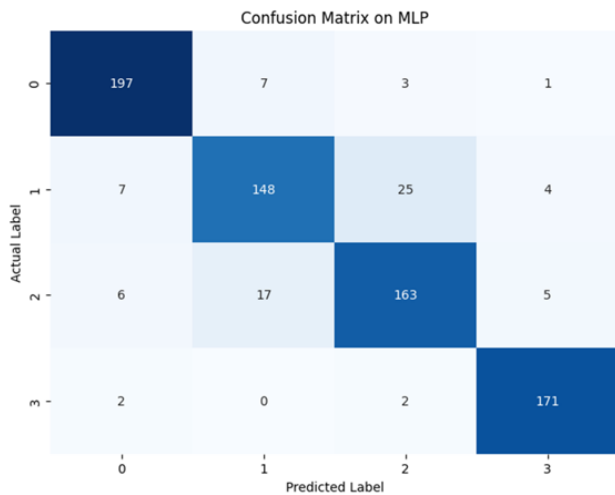


Fig. 8. Confusion Matrix – MLP

7) *Random Forest*: Random Forest outperformed the majority of traditional machine learning models. Stable generalization and high accuracy. Very little misclassification. Excellent results because of the combination of ensemble features. In conclusion, this is one of the most effective ML methods.

8) *XGBoost*: Among ML models, XGBoost demonstrated strong categorization across all tumour types, achieving the greatest accuracy. successfully manages non-linearities. Able to withstand overfitting. Excellent performance on high-dimensional feature sets. In conclusion, this machine learning model is the best overall and is very dependable for classifying tumours.

B. Convolutional Neural Network (CNN) Efficiency

When compared to all ML models, the CNN performed better, accomplishing: Training Precision: around 99%. Accuracy of Validation: 93%. Minimal overfitting and stable loss convergence.

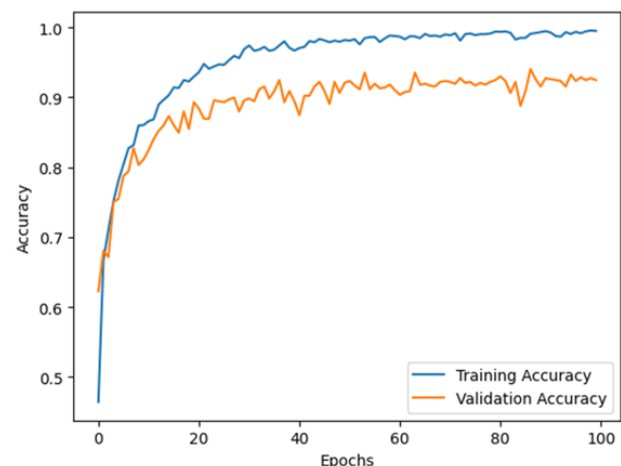


Fig. 10. CNN Training and Validation Accuracy

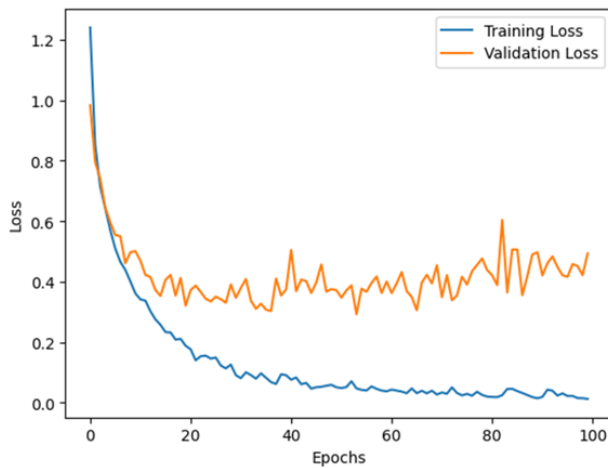


Fig. 11. CNN Training and Validation Loss

1) *Why CNN Did Better Than ML:* Multi-level characteristics (edges → textures → forms → tumour areas) are automatically learned. more resilient to changes in tumour form and MRI orientation. removes the need for hand-crafted, manual feature engineering.

2) *CNN Loss and Accuracy Trends:* For validation, accuracy grew gradually and reached a plateau at about 93%. Validation loss stabilized without divergence, whereas training loss became closer to zero. shows a highly predictive model that is well-generalized. Conclusion: Because CNN can directly learn hierarchical spatial patterns from MRI pictures, it obviously outperforms all ML techniques.

C. Model Performance Summary

TABLE I
MODEL PERFORMANCE SUMMARY

Naïve Bayes	Poor; cannot model image complexity
KNN	Moderate; sensitive to noise
Logistic Regression	Moderate; limited by linearity
Decision Tree	Good but overfitted
SVM	Very good; strong generalization
MLP	Very good; learns non-linear patterns
Random Forest	Excellent; robust ensemble
XGBoost	Outstanding; best ML performer
CNN	93% accuracy, best overall

The findings unequivocally show that deep learning outperforms conventional machine learning for MRI-based tumour categorisation. CNN is better at extracting and interpreting deep spatial patterns than Random Forest and XGBoost, despite their superior performance with designed features.

CNN's potential for incorporation into clinical decision-support systems is highlighted by its high accuracy (93%). CNN performance can reach radiologist-level accuracy with enough data and transfer learning.

All things considered, this comparison analysis confirms that CNNs are the best option for automated, multi-class brain tumour classification, but ML models can be quick, less complicated substitutes.

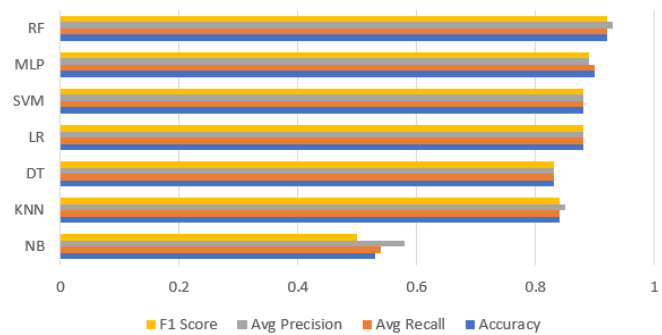


Fig. 12. Performance comparison of ML models showing Accuracy, Precision, Recall, and F1 Score across NB, KNN, DT, LR, SVM, MLP, and RF.

TABLE II
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	Avg Recall	Avg Precision	F1 Score
Naïve Bayes	0.53	0.54	0.58	0.50
K-Nearest Neighbors	0.84	0.84	0.85	0.84
Decision Tree	0.83	0.83	0.83	0.83
Logistic Regression	0.88	0.88	0.88	0.88
Support Vector Machine	0.88	0.88	0.88	0.88
Multilayer Perceptron	0.90	0.90	0.89	0.89
Random Forest	0.92	0.92	0.93	0.92

VII. DISCUSSION

Random Forest and XGBoost achieved the highest performance among ML models, due to their ability to combine multiple weak learners. Classical classifiers like Naïve Bayes and Logistic Regression performed poorly due to MRI complexity.

The CNN model outperformed all ML models with a validation accuracy of 93%, thanks to its ability to extract deep spatial features without manual engineering. CNN loss curves showed stable convergence, indicating effective generalization.

Overall, CNN is the most robust approach, while XGBoost provides a fast and lightweight alternative.

VIII. CONCLUSION

Eight machine learning techniques and a convolutional neural network (CNN) for multi-class brain tumour classification using MRI data were thoroughly compared in this study. On a dataset of 3,788 MRI scans, a bespoke CNN architecture was compared against traditional machine learning models such as Naïve Bayes, KNN, Logistic Regression, Decision Tree, SVM, MLP, Random Forest, and XGBoost. The findings showed that even though a number of machine learning models—Random Forest and XGBoost in particular—performed well, they were still constrained by their reliance on manually created features and their incapacity to completely capture the intricate spatial patterns found in MRI scans. On the other hand, because of its deep feature-learning capabilities, the CNN model demonstrated greater generalization and achieved a validation accuracy of 93%, outperforming all ML techniques. In order to distinguish between gliomas, meningiomas, pituitary tumours,

and normal brain scans more accurately, the CNN successfully learnt hierarchical spatial and textural properties straight from raw MRI images.

Overall, the results show that the most dependable and scalable method for automated brain tumour classification is deep learning, especially CNN-based architectures. CNN-based systems offer great potential to assist radiologists in early diagnosis and improve clinical decision-making in actual medical settings with additional advancements like bigger datasets, transfer learning, and integration of sophisticated architectures like ResNet or EfficientNet.

IX. ACKNOWLEDGMENT

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